Ben-Gurion University of the Negev

Faculty of Engineering

Department of Mechanical Engineering

Introduction to Image Processing Course – First Assignment

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1. Environment Setup and Preparations

1.1. Library versions

```
print('pandas version: '+ pd.__version__)
print('numpy version: '+np.__version__)
print('tensorflow version: '+tf.__version__)
print('scikit-image version: '+ski.__version__)
```
[2] ✓ 0.0s Python

```
pandas version: 2.0.3
numpy version: 1.26.0
tensorflow version: 2.10.0
scikit-image version: 0.19.3
```

1.2. Environment check

```
conda_env = os.environ.get('CONDA_DEFAULT_ENV')
print(f"current Conda environment: '{conda_env}'.")
```
[36] Python

```
current Conda environment: 'image_processing_env'.
```

1.3. Images preparation for the course

The images that will be using for this assignment are shown below.

1.4. Images sizes comparison

```
Selfie:
  Size (Height x Width x Channels): (497, 319, 3)
  Data Type: uint8

Friends:
  Size (Height x Width x Channels): (1551, 1060, 3)
  Data Type: uint8

Street:
  Size (Height x Width x Channels): (1600, 900, 3)
  Data Type: uint8

Animal:
  Size (Height x Width x Channels): (1338, 718, 3)
  Data Type: uint8

Words:
  Size (Height x Width x Channels): (648, 1600, 3)
  Data Type: uint8
```

1.5. Image Formats and Color Schemes

This part explores the advantages and disadvantages of different color schemes (RGB, CMYK, HEX, Grayscale, HSL) and file formats (TIFF, JPEG, PNG, GIF).

1.5.1. RGB (Red, Green, Blue)

RGB is ideal for digital screens and online content due to its wide color gamut and vibrant display. However, it is not suitable for print as it does not directly translate to physical colors.

1.5.2. CMYK (Cyan, Magenta, Yellow, Key/Black)

CMYK is specifically designed for print, ensuring accurate color reproduction on paper. Its primary limitation is a narrower color range compared to RGB, and it requires conversion for digital use.

1.5.3. Grayscale

Grayscale simplifies visuals, reduces distractions, and conserves ink in print. Its downside is reduced engagement and limited differentiation between elements.

1.5.4. HSL (Hue, Saturation, Lightness)

HSL is intuitive for adjusting colors in design and is easier to manipulate for specific shades and tones. However, it may require conversion for use in some applications.

1.5.5. TIFF (Tagged Image File Format)

TIFF is a high-quality, lossless format suitable for professional use, particularly in editing and archiving. Its major drawbacks are large file sizes and limited compatibility with some applications, such as web or mobile applications.

1.5.6. JPEG (Joint Photographic Experts Group)

JPEG is a widely supported, compressed format that is ideal for web use, balancing quality and file size. However, its lossy compression can degrade quality with repeated saves making this format a trade-off between quality and efficiency.

1.5.7. PNG (Portable Network Graphics)

PNG offers lossless compression and supports transparency, making it ideal for web graphics and logos. Its downside is a larger file size compared to JPEG for complex images.

1.5.8. GIF (Graphics Interchange Format)

GIF supports simple animations and transparency, making it lightweight and suitable for small graphics. However, it has a limited color palette (256 colors) making it unsuitable for high-quality images.

	RGB	Grayscale	CYMK	HSL
.tiff				
.jpeg				
.gif			-	
.png			-	

As seen in the table above, CYMK does not support all file formats (.gif and .png in particular). Another detail to note is that the images converted to .gif file formats look slightly different due to a smaller color pallet.

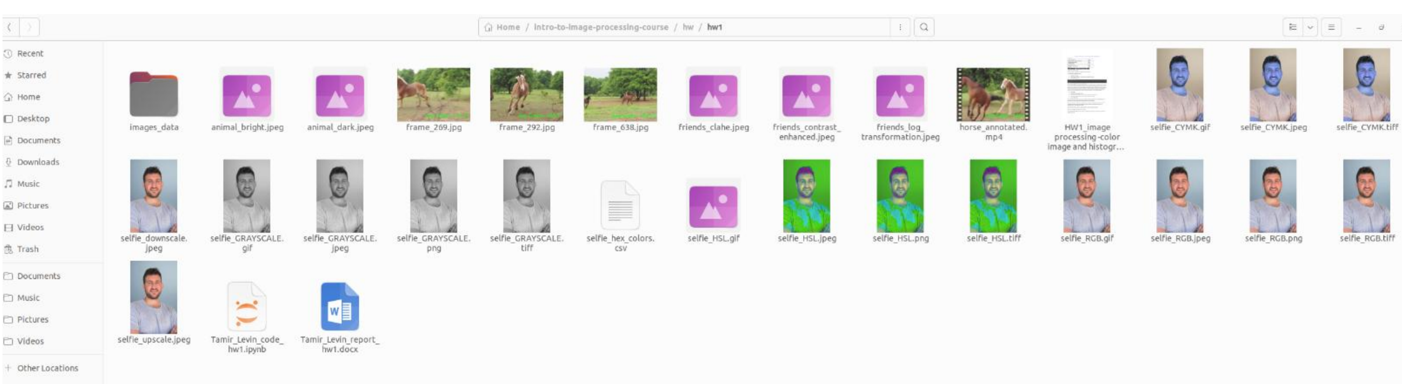
1.5.9. HEX (Hexadecimal)

HEX is the standard for web design, ensuring consistent color across platforms. However, it is limited to digital applications and can be less intuitive to interpret compared to RGB or HSL.

```
Hex color codes saved to selfie_hex_colors.csv
[['#7c91a2' '#7c91a2' '#7c91a2' ... '#a7bac8' '#a8bbc9' '#a9bcca']
 ['#7c91a2' '#7c91a2' '#7d92a3' ... '#a6b9c7' '#a7bac8' '#a8bbc9']
 ['#7c91a2' '#7d92a3' '#7d92a3' ... '#a6b9c7' '#a6b9c7' '#a6b9c7']
 ...
 ['#da8f78' '#da8f78' '#da8f78' ... '#804b3b' '#7e4937' '#804837']
 ['#dc9079' '#db9079' '#dc9079' ... '#7b4636' '#7a4533' '#7d4836']
 ['#dd8f79' '#db8f78' '#dc8e78' ... '#814c3c' '#834e3e' '#855040']]
```

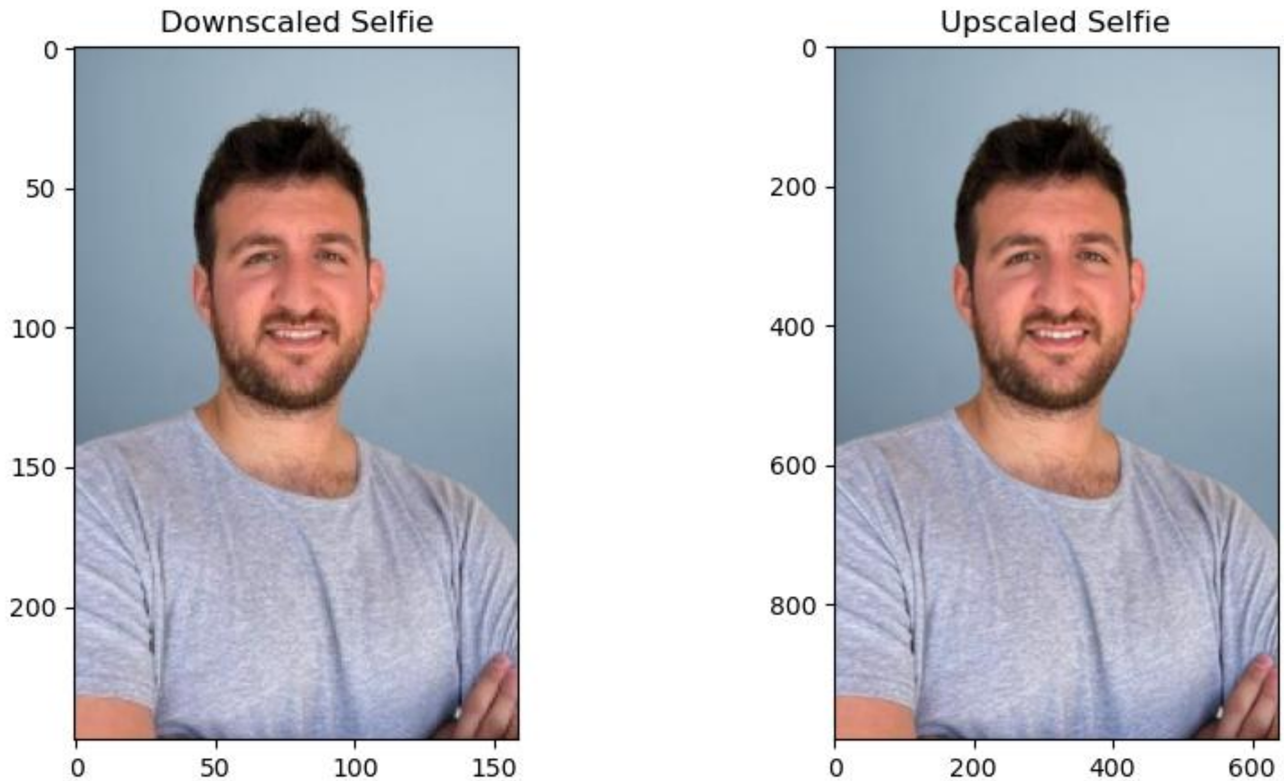
The Hex format image can be then converted to .csv file to represent the image as a table.

1.6. Screenshot of the folder with files in different image and color formats



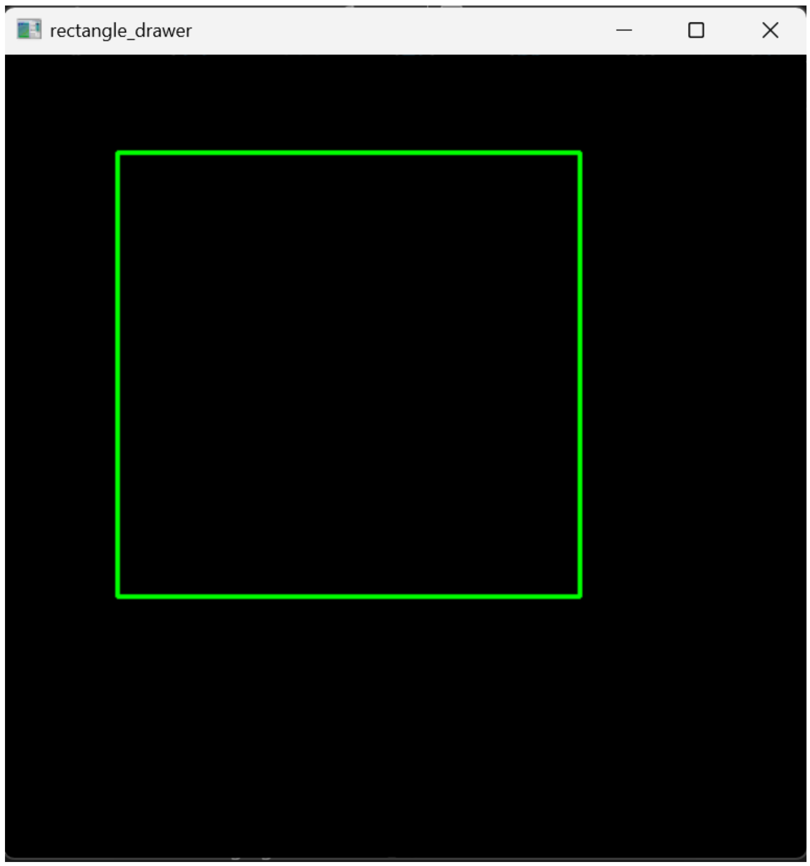
2. Upscaling and Downscaling

In the pictures below we can see a comparison of downscaled and upscaled versions of the same image. The upscaled image size is 110KB while the downscaled image size is only 15.5KB (both in .jpg format). It is also can be seen that the upscaled image is more blurred than the downscaled image, which makes sense.



3. Shape Drawing Results

In the image below we can see a demo of a python script that been made to allow to draw a rectangle anywhere on an image based on the user's mouse coordinates. The program can also be reset by pressing 'r' on the keyboard or exit by pressing 'esc'.



4. Video Annotation Results

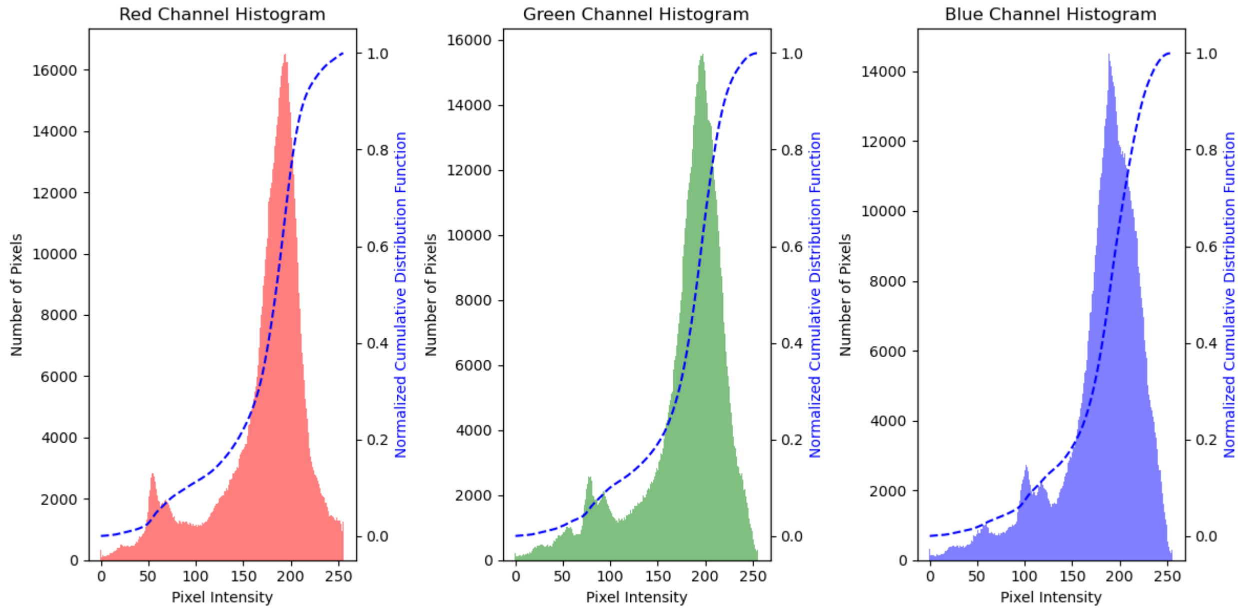
In the images below we can see 3 random frames from video that been made of a python script. The script allows you to read a given video and write over each frame the average green-ban value in the right-bottom corner as seen. We can see that in the middle image, where the brown horse is the furthest from the camera, which means the greenest background, the average green value is the highest.

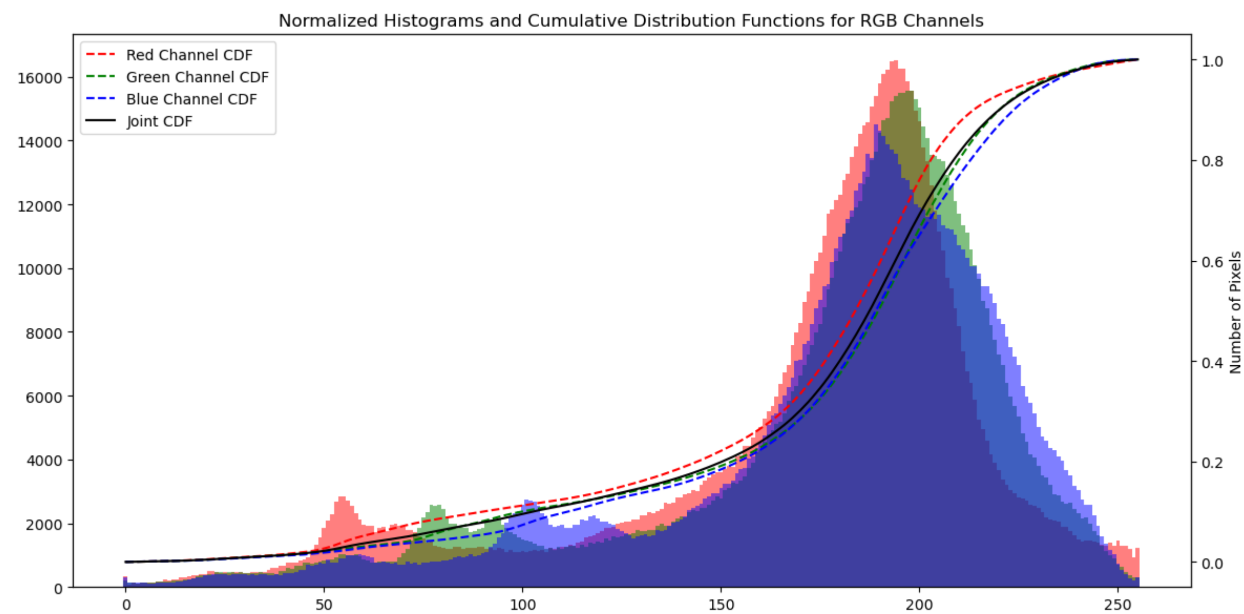
5. Image Histogram Analysis

In this part the ‘animal’ image has been used to generate histogram analysis for each color channel of the image and a cumulative distribution function. The original image that been used is shown below:

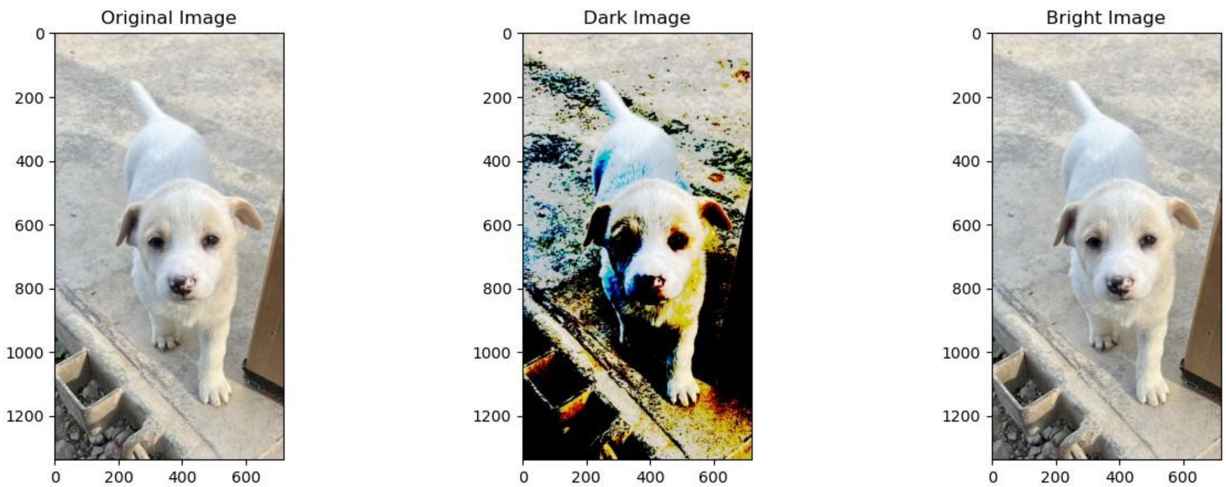
As we can see, the colors in this image are quite bright so we can expect the histogram to be right leaning towards the higher values. We can also expect all the 3 different histograms to have around the same distribution since there is no one dominant color in this image (red, green or blue).

The generated histograms can be seen below:





In the images below we can see the original image compared to darkened and brightened images made using a threshold. For the darkened image threshold value is set to `180` and the brightened image threshold value is set to `10`.



For this part I chose to use the ‘friends’ image. In the images below we can see the original image compared to one that a Log transformed was applied onto.

Original Image

Image after Log Transformation

In the images below we can see the original image compared to one that a Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied onto.

Original Image

CLAHE Processed Image

In the images below we can see the original image compared to one that a Contrast enhancement by Histogram equalization was applied onto.

Original Image

Contrast Enhanced Image